CyBOK

Memory analysis

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Agenda

- Intro to theory
  - Data in memory
  - Malware structures
  - Memory analysis with Volatility
- Volatility demo
  - Set up the environment
  - Quick start

Litterature note

- This lecture is based on the following CyBOK knowledge areas:
  - Forensics
  - Malware & Attack Technologies

Data in memory

- Running memory holds information relating to what the computer is currently up to

- Volatile
  - Temporary
  - Removed on power-cycle

- Recent
  - Put there since last boot

Data in memory – properties

- True form
  - Unencrypted
  - Unobfuscated

- Memory-only data
  - Memory resident malware
  - Temporary data such as passwords

Data in memory – special cases

- SWAP-partitions
- Pagefile/hiberfile
- Memory actually fades, rather than is directly removed, on power-off

Data in memory

- Memory is an array of pages

- Processes are not assigned pages in physical memory but a virtual memory space
  - Size equal to physical memory
  - Easier memory allocation for the process
  - A page table maps virtual memory to physical memory

Data in memory – simplified paging

ADDRESS	Process A	Process B	Process C	Page Table	Physical memory	ADDRESS
1				B:1-5		1
2				A:6-8		2
3				A:11-12		3
4				C:9-10		4
5				C:12-15		5
6						6
7						7
8						8
9						9
10						10
11						11
12						12
13						13
14						14
15						15
16						16
17						17
18						18
19						19
20						20
21						21
22						22
23						23
24						24
25						25

Data in memory

- Different operating systems handle memory a bit differently
- Paging scheme
- Address space layout randomization (ASLR)
  - Some address space positions are randomized to make it hard for an attacker to guess the memory location of important processes

Data in memory

- What are the forensic purposes of memory analysis?
  - Recent nature of data
  - True form of data
- Memory is where the instructions to be executed are stored
  - Malware using evasion techniques
  - Bad behaviour is manifested
- True form of data
  - Encrypted data in plain text
  - Nuggets of temporary information
  - Volatile data

REFLECT!

- Use one minute to write down one-liners which describe what you just learned – come up with as many as you can.
- Join up in pairs or groups and compare your lists for another minute.
- Questions so far?

Moving on

Malware

- Malware is any malicious program

- Great deal of diversity in
  - What they do
  - How they spread
  - How they hide

Malware – what they do

- Encrypt data
- Create a backdoor
- Exfiltrate information
- Slow down computer
- Destroy information
- Enlist device in bot-net
- ….

Malware – how they spread

- Attachments
- Drive-by-download
- USB-stick
- Using software vulnerabilities
- Manually
- …

Malware – how they hide

- Encryption
- Polymorphism or Metamorphism
- Memory-resident (file-less)
- ….

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Moving on

Memory analysis with Volatility

- Memory analysis is about analyzing memory with the intent of finding something
- On-line analysis of a running computer
- Off-line analysis of a memory dump

Memory analysis with Volatility

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- Off-line analysis is better when possible
  - No need to allow infected machines to run
  - Less time-sensitive because a machine state is stored indefinitely
  - Complies with forensic best practice
- Requires memory to be dumped with a tool or cold-boot procedure
- Requires a tool that can understand the content of a memory dump

Memory analysis with Volatility

- Volatility is a command-line tool for memory analysis

- Two versions
  - 2.6: discontinued but still with many useful functions
  - 3: newer but less extensive in terms of functionality

- There is also a GUI tool called volatility workbench

Memory analysis with Volatility

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- Malware is an anomaly
- Finding an anomaly means knowing what is normal and find deviations from that
- Being a good memory analyst requires research and experience

Memory analysis with Volatility

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- A helpful starting point is to consider what malware may want to do
  - In an incident handling process this would be manifested as “the incident”
  - Find what is causing the anomalous behaviour

- Some starters
  - Spawn processes
  - Irregular process behavior (start/stop)
  - Address strange memory areas
  - Spawn network sockets
  - Issue command line arguments
  - Store data in the registry (Windows)
  - Encrypt

Memory analysis with Volatility

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- Use the internet!
  - Research the incident at hand
  - Use sites such as Virustotal for confirmation and review
- Indicators of compromise
  - “manifests” of known malware
  - Can be used to search for bad behavior
  - Or specific malware
  - Some tools can do this automatically
- Let’s demo.

Volatility– set up

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- Volatility 2.6 installation in Windows
  - Go to https://www.volatilityfoundation.org/26
  - Download Volatility 2.6 standalone executable
  - Open a PowerShell terminal and navigate to the folder containing Volatility
  - List of commands: https://github.com/volatilityfoundation/volatility/wiki/Command-Reference

- Volatility 3 installation in Windows (You need python 3)
  - Download AND UNPACK Volatility 3
    - From web page: https://www.volatilityfoundation.org/3
    - Clone from git (ensures latest version): git clone https://github.com/volatilityfoundation/volatility3.git
  - Install some dependencies
    - *Python -m pip install pefiles*
  - Open a PowerShell terminal and navigate to the Volatility folder containing “vol.py”
  - You can now run Volatility by issuing the command *python vol.py*

Volatility 2.6

- Framework with a set of modules

- Executed by running the executable from command line
  - `.\volatility_2.6_win64_standalone.exe -f` *`memdumpfilename`* `--profile=”`*`profilename`*`”` *`modulename`*

- Verify by asking for help
  - `.\volatility_2.6_win64_standalone.exe -h`

Volatility 3

- Script that calls other scripts (modules)

- Executed by running the executable from command line
  - `Python vol.py -f` *`memdumpfilename modulename`*

- Verify by asking for help
  - `Python vol.py -h`

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Demo time – quick start

- Let’s begin with a short demo before you explore on your own.
- The demo is available in the compendium

Thank you for today!